

AgriVue

Smarter Decisions, Every Day, From Seed to Sale.



Yuva Yodha Energy Tech Hackathon

An AI-powered farm assistant that gives farmers personalized daily plans, crop monitoring, and harvest-to-market guidance, all based on their own field, crop, and weather.

CHALLENGE 01

Sustainable Agriculture

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Problem Statement →

The agricultural industry of India depends on small farmers, who make most of their decisions without any immediate expert advice or help. According to the Agriculture Census 2015-16, small and marginal farms, which include those having less than 2 hectares of land, comprise more than 86% of the total holdings, about 125 million farms, and the average farm size has shrunk from 1.15 to 1.08 hectares [1]. Thus, there are millions of small fields in which a single mistake made by the farmer may lead to financial loss.

1. Wrong timing at sowing. Yield might depend on the sowing date alone. For example, according to a study conducted in the rice-wheat belt of Eastern India and published in Nature Foods, every day of delayed sowing resulted in a loss of yield potential of around 25 kg/ha until 20 November, and the rate of yield loss more than doubled to 51 kg/ha/day afterward [2]. Early-sown farmers in this study recorded yields that were around 1 tonne per hectare higher compared to the yield from late-sown farmers, which amounts to an increase of 36% [2]. Despite this fact, late sowing was observed in 84% of the 5,766 surveyed cases [2].

2. Water wasted through guesswork irrigation. Approximately 80-85% of India's irrigation practices involve flood irrigation where water-use efficiency is as low as 30-40%, meaning that the majority of water pumped ends up not being used by the plant roots [4]. This is significant because agriculture alone consumes 89% of India's groundwater consumption and 11% of India's groundwater assessment units have been categorized as overexploited and 3.05% as critical [3]. In total, India utilizes 2 to 3 times more water than China, Brazil, or the US to cultivate food crops.

3. Energy burned by unscheduled pumping. Indian Irrigation is based predominantly on groundwater drawn by approximately 9 million diesel and 21 million electric pumps [6]. Up to 20% of total energy consumption in India is for agriculture, and most of this is for irrigation, while in some states it is even up to 30-50% [7]. When power is provided for free during the night, farmers use pumps all night long, which leads to over irrigation of the land, depletion of underground water resources, and waste of energy [7].

4. Pests and diseases caught too late. About 15% to 25% of India's yearly yield gets destroyed by different kinds of pests [8]; according to the government, the number stands at about 30% [9]. The losses in major vegetable crops vary between 46% to 100%, resulting from insects, and 5% to 90%, caused by diseases [10]. However, all this is preventable if only pest issues are detected in time; however, no one does any scouting on the farmsteads, and therefore all the damage becomes obvious when there are visible signs across the whole field.

5. Advice that doesn't reach the farmer. Guidance is scarce. In 2018, nearly 48.7% of farmer households availed themselves of technical help related to agriculture, which reduced to 42.2% in 2019 [11]. In the public sector too, an examination of the NSS data reveals that ATMA managed to cover merely 1% of farmers [12]. The majority of farmers tend to depend on input retailers, neighbours, or tradition, none of which takes into account their particular soil, crop condition, and weather conditions.

6. Value lost after harvest. However, even a good crop loses its value after being harvested. The study conducted by NABCONS 2022 for the Ministry of Food Processing Industries revealed that each year food worth ₹1.53 lakh crore is lost due to post-harvest losses [13], accounting for 2.35% of the country's GDP [15]. It varies from 3.89% – 5.92% of losses in cereals, 5.65% – 6.74% in pulses, to 6.02% – 15.05% in fruits and 4.87% – 11.61% in vegetables [14]. Lacking any recommendations regarding storage time, rate of degradation and optimal timing for selling produce, farmers usually sell their goods after harvesting at low prices or store them inadequately and lose both quantity and quality.

The core gap. These are not six different problems. They have one common origin: farmers make dozens of time-bound decisions on a seasonal basis (what days to plant, how many hours to operate the pump, whether to spray the leaf spots, what days to harvest), relying not on their farm's data, but on tradition and hearsay. Each mistake costs several percentage points. All added together, it leads to low yields, inefficient use of water and electricity, higher cost of inputs, and money loss during post-harvesting processing. The need for a cheap and easy-to-use system which would turn data about location, crops, weather, and field imagery into proper actions at the right moments – this is the problem AgriVue solves.

Solution →

AgriVue is an agribusiness AI solution that provides customized daily planning to each individual smallholder farmer for his or her own farm from planting to selling. The farmer makes a simple profile (of location, type of crop, area of crop, date of sowing, irrigation technique), and AgriVue analyzes this information along with weather and crops data to determine the suitability of the farming process and to identify the right windows for sowing and harvesting. After that, AgriVue forms a daily task list, which includes the exact amount of watering, time of irrigation and pump operation, necessary actions for crop care and pest control, and types of input products used in agriculture. Every week, the farmer has to upload two pictures, one close-up and another overview of the entire field, which will be analyzed by AgriVue to identify any pests, diseases, or stress on plants, and monitor plant growth. Savings on water usage, energy consumption, inputs, and labor are recorded and presented to the farmer, while an AI assistant responds to queries in either text or voice format in the farmer's native language, based on the particular farm. Once the harvest is done, AgriVue helps in storing the produce and forecasts quality degradation and timing of sale.

How does the solution work ?

The way AgriVue functions is a circular process, not a one-time analysis. A farmer gives a certain amount of information at once, and AgriVue processes it into an entire growing season's plan that gets better and better as more weather-related and image-based information is provided. As shown in Fig. 1, the system is organised into five layers: data sources, data processing, AI/ML models, the farmer-facing application, and a storage and integration layer. New weather data and weekly field photos continuously feed back into these layers, so the plan keeps updating through the season.

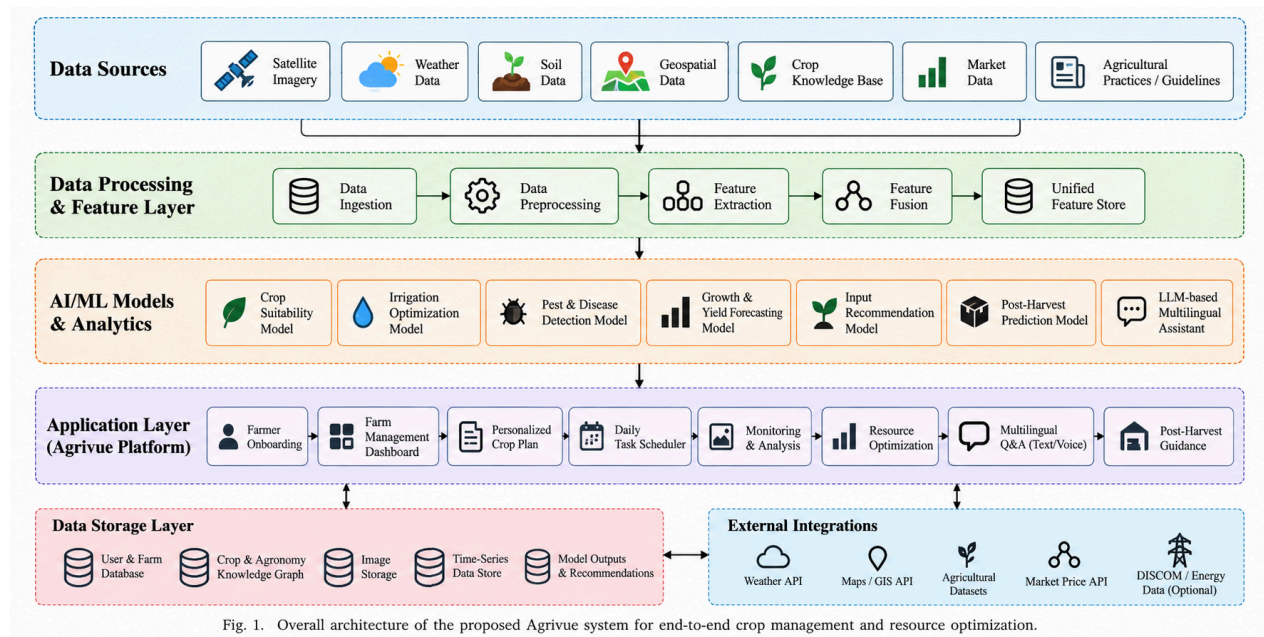


Fig. 1. Overall architecture of the proposed AgriVue system for end-to-end crop management and resource optimization.

Input Data. The input data for AgriVue is based on three factors. First, the farmer's details are provided at the time of signup: place of agriculture, crop(s), land size, sowing time, current growth stage, type of irrigation, and farming objective (such as high yield or minimum cost). Second, this data is supplemented by external data related to the location, which consists of short-term weather forecast, precipitation levels, temperature and humidity conditions, soil data from the region, standard crop calendars and water requirement, and the mandi price trends. The third factor is the images uploaded by farmers each week of the field condition.

Crop analysis. If a farmer is to register with AgriVue, the first step involved will be the selection of the crop that is to be planted. In doing so, AgriVue analyzes whether the selected crop is suitable to the area depending on the weather and soil characteristics. The optimum period for planting is determined, together with when the crop will pass into different growth stages and reach maturity stage.

Daily optimized plan. Every day, AgriVue generates an easy to-do list based on the crop plan and the current weather conditions. Regarding irrigation, it figures out the water requirements of the crop at this growth stage, deducts rainfall amounts and adjusts for the mode of irrigation to come up with clear guidelines that indicate if irrigation is required, for how many hours to turn on the pump and when to do so – be it in the early morning hours to reduce evaporation losses or during power-supply hours. Along with that, there is an instruction on crop maintenance activities at each stage of growth including weed eradication, fertilizer top-dressing or earthing-up, together with pest or disease preventive activities due to hazardous weather conditions (high humidity, etc.). There are also input requirements and what to refrain from.

Weekly crop monitoring. Every week, two photos are uploaded by the farmer. The first one of either leaves or plant is analyzed for any signs of pest attack, disease or lack of water and nutrients. The second photo is of the entire field and is analyzed to determine the level of canopy cover and development of the crop. The system will then compare each week's results to those from previous weeks as well as to the expected results for the particular stage of growth. If there is any deviation, the farmer is immediately alerted on the findings, severity and course of action.

Forecasting and early warnings. With the use of the crop plan, weather, and weekly tracking, AgriVue helps the farmer understand where his crop is at currently, how it will grow in the upcoming weeks, the time frame for harvesting and future jobs to do. In case the growth of the crop is behind schedule or there is a potential threat of heat stress, heavy rains or pests attacks, the farmer will be notified about it.

Resource optimization. All recommendations are measured against what a farmer would otherwise be using. AgriVue documents the amount of water used, pumping hours, energy or diesel used, inputs used, and labor used, and then provides the farmer with a continuous estimation of how much they have saved against their usual practices. The savings are in terms of rupees and liters, which helps increase trust.

AI assistant. The farmer may at any time ask a question either through typing or speaking in his/her own language: “Why do my leaves turn yellow?” Or “Should I irrigate before rain falls tomorrow?” Since the AgriVue application has already got the details regarding what kind of crop the farmer is growing and at which stage and the climatic conditions prevailing, etc., the answer would be relevant to the farmer's field itself.

Post-harvest guidance. As the time for harvest draws near, the emphasis for AgriVue moves towards preserving the value of the crop. This involves advising the farmer on how to dry and store the crop, estimating the period during which the crop can be stored without any reduction in its quality, and incorporating this information with the local market dynamics in order to recommend the optimal time for sale.

Key Assumptions:

Agrivue is premised on some practical assumptions. Farmers or a member from within the household has access to an entry-level Android mobile phone with a camera, which is becoming more common among farmers in rural India. Access to the internet might not be always consistent; hence the plan for the day will be downloaded ahead of time and work offline until internet access becomes available. Information about reliable weather forecasts and regional soil and crop information is readily available at the district or village level. Also, the farmers are ready to dedicate some time each week to take pictures and check their plan for the day. Lastly, field-level partnerships from FPOs, KVKs, or agri-entrepreneurs will assist in the initial stages.

Why It Suits Indian Smallholder Conditions:

AgriVue is designed keeping the constraints of an individual farmer having an acre or two in mind and not that of a big commercial farm. AgriVue does not need any sensors or drones and requires no extra investment, just a phone which is already owned by the farmer. Voice support along with regional languages help in making it accessible for the illiterate farmers, while the suggestions are made in the form of practical actions and not complex scientific reports. AgriVue schedules pumping keeping in mind the non-uniformity in power supply which is a daily constraint of all irrigating farmers. As all the suggestions are related to the specific field, crop and weather of the farmer, AgriVue fulfills the gap created due to inadequate extension service and generic advisory service. Being water, electricity, and input cost-saving solutions makes it suitable for farmers operating within narrow profit margins where every extra irrigation or spray can make a difference.

Expected Impact:

On an individual farm basis, the objective for AgriVue is to reduce the amount of water for irrigation and pump operation through need-based scheduling, save on fertilizers and pesticides from wasteful practices, and ensure higher yields by appropriate scheduling of sowings and pest/disease management at the initial stages. Energy saving is also directly related here, as a pump of 5 HP will consume about 3.7 kW of power, and not operating even one additional hour of pumping during irrigation will result in saving of around 370 units per farmer in 100 days. Overall, for 1 lakh farmers, this will be 37 GWh per season along with savings on diesel consumption, less burden on DISCOMs, and less extraction of groundwater. Better storage and selling of produce can help to retain the lost income of ₹1.53 lakh crore post-harvest every year. AgriVue would initially start with a pilot program for one or two crops in one district only.

References

- [1] Department of Agriculture, Cooperation & Farmers Welfare (2020). *All India Report on Agriculture Census 2015-16*. Ministry of Agriculture & Farmers Welfare, Government of India.
- [2] Nayak, H.S. et al. (2022). Time management governs climate resilience and productivity in the coupled rice–wheat cropping systems of eastern India. *Nature Food*, 3, 542–551. [Link](#)
- [3] Central Ground Water Board (2024). *Dynamic Ground Water Resources Assessment of India, 2024*. Ministry of Jal Shakti, Government of India. [Link](#)
- [4] Irrigation in India: Status, challenges and options. *ResearchGate*. [Link](#)
- [5] India Water Portal (2024). Micro-irrigation and groundwater use. [Link](#)
- [6] International Energy Agency (2022), estimated stock of irrigation pumps in India, as reported in *Frontiers in Environmental Science* (2024). [Link](#)
- [7] World Bank Blogs. Will the Sun God answer poor farmers' prayers or make things worse? [Link](#)
- [8] Dhaliwal, G.S., Jindal, V. & Mohindru, B. (2015). Crop losses due to insect pests: Global and Indian scenario. *Indian Journal of Entomology*, 77(2), 165–168.
- [9] SciDev.Net. Climate, pests and pollution fuel crop losses across India. [Link](#)
- [10] Crop Loss Assessment in India: Past Experiences and Future Strategies. *ResearchGate*. [Link](#)
- [11] National Statistical Office (2021). *Situation Assessment of Agricultural Households and Land and Livestock Holdings of Households in Rural India, 2019* (NSS 77th Round). Ministry of Statistics and Programme Implementation, Government of India.
- [12] Access to agricultural extension sources among Indian farmers: evidence from NSS 77th Round. *Frontiers in Sustainable Food Systems* (2024). [Link](#)
- [13] NABARD Consultancy Services (NABCONS) (2022). *Study to Determine Post-Harvest Losses of Agri-Produces in India*. Ministry of Food Processing Industries, Government of India.
- [14] Ministry of Food Processing Industries (2024). Rajya Sabha Unstarred Question No. 2145, answered 9 August 2024. [Link](#)
- [15] Factly (2026). From Guava to Milk: What Food Loss Reveals About India's Supply Chains. [Link](#)